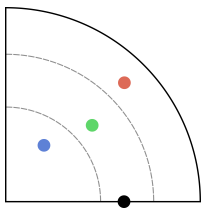


polar_2 → base 07.05.2026, 16:44

Page 01

INTRO

IN01

Introduction

Study Title

Enhanced Summary Polar Diagrams for Model Comparison

Purpose of the Study

This study evaluates a new enhanced implementation of the *polar-diagrams* Python library. The implementation consists of a suite of visualization techniques, including overview+detail, aggregation, filtering, and Cartesian linking. The study examines whether this combination of visualization techniques enables users to efficiently complete the tasks for which the summary polar diagrams were designed. Since the summary polar diagrams themselves do not have any visual differences, the study focuses on a single type of such a diagram - the Scaled Mutual Information diagram.

What You Will Be Asked to Do

As a participant, you will be asked to solve analytical tasks using the enhanced implementation of the summary polar diagrams. The study has four distinct parts:

1. **Introduction** - This page introduces you to this study. It is the page you are currently on.
2. **General Questions** - This page presents a couple of sociodemographic questions, along with a few questions relevant to the study.
3. **Tutorial-Task Combinations** - This section introduces the summary polar diagrams and describes the new visual techniques implemented on top of them. Each tutorial section is followed by multiple pages that present different tasks for you to complete.
4. **Conclusion** - The final section implements a system usability test and collects user feedback on the questionnaire and the system.

Duration

The study will take approximately **20 minutes** to complete.

Risks and Benefits

Participation in this study carries no known risks. The benefits include contributing to cutting-edge research and gaining a better understanding of the summary polar diagrams.

Voluntary Participation

Your participation in this study is entirely voluntary. You may choose to withdraw at any time without any negative consequences.

Confidentiality

We will not collect any personal identifying information during this study. The sociodemographic data gathered will be used solely for research purposes and kept strictly confidential. All data will be stored securely and anonymously, in compliance with GDPR regulations.

Data We Collect

We will collect some general demographic details from you, including your age range, gender, education level, years of experience, and occupation category. This information will not be used to identify you individually. Additionally, we will ask you to self-assess your ability to understand and interpret visualizations, evaluate the overall system, and provide optional feedback. We will also measure the time it takes you to complete each analytical task.

Use of Data

The anonymized data collected in this study will be used only for academic research purposes.

Questions or Concerns

If you have any questions or concerns about this study, please contact: Aleksandar Anžel (AnzelA@rki.de).

Consent Declaration

By participating in this study, you are indicating that:

1. You have read and understood the information provided above.
2. You voluntarily agree to participate.
3. You understand that you can withdraw from the study at any time without penalty.
4. You are at least 18 years of age.

To indicate your consent, please **check the box below** and click **Next**.

IN02

☐ I agree to participate in this study.

SD17

General Questions

SD05

1. What is your gender?

- Woman
- Man
- Transgender
- Non-binary/non-conforming
- Prefer not to respond

SD03

2. How old are you?

- 20 to 30 years old
- 30 to 40 years old
- 40 to 50 years old
- 50 to 60 years old
- 60 to 70 years old

SD12

3. What is the highest degree or level of school you have completed? If currently enrolled, highest degree received.

- No schooling completed
- Nursery school to 8th grade
- Some high school, no diploma
- High school graduate, diploma or the equivalent (for example: GED)
- Some college credit, no degree
- Trade/technical/vocational training
- Associate degree
- Bachelor’s degree
- Master’s degree
- Professional degree
- Doctorate degree
- Habilitation
- Other degree

4. What is your field of expertise? Multiple choices are allowed. SD13

- [ ] Business and Management
- [ ] Engineering and Technology
- [ ] Healthcare and Medicine
- [ ] Education and Teaching
- [ ] Arts and Design
- [ ] Law and Legal Services
- [ ] Finance and Economics
- [ ] Information Technology
- [ ] Sciences (Physical, Life, and Social)
- [ ] Media and Communications
- [ ] Agriculture and Environmental Studies
- [ ] Public Service and Administration
- [ ] Hospitality and Tourism
- [ ] Manufacturing and Construction
- [ ] Social Services and Counseling

5. How many years of experience in the selected fields do you have? SD14

- ○ < 5 years of experience
- ○ 5 – 10 years of experience
- ○ 10 – 15 years of experience
- ○ 15 – 20 years of experience
- ○ 20 – 25 years of experience
- ○ 25 – 30 years of experience
- ○ 30 – 35 years of experience
- ○ > 35 years of experience

6. Please answer the following subjective graph literacy test: SD18

	not at all good					extremely good
How good are you at working with bar charts?	○	○	○	○	○	○
How good are you at working with line plots?	○	○	○	○	○	○
How good are you at working with pies?	○	○	○	○	○	○
How good are you at inferring the size of a bar in a bar chart?	○	○	○	○	○	○
How good are you at determining the difference between 2 bars in a bar chart?	○	○	○	○	○	○

SD15

7. Do you have experience with charts in the polar coordinate system?

- ◯ Yes
- ◯ No

SD16

8. Do you have experience with summary polar diagrams such as the Taylor or the Mutual Information diagram?

- ◯ Yes
- ◯ No

NA05

Next Steps

In the next section, you'll go through a series of tutorials followed by two tasks. The tutorials will use different data than the tasks, so you'll need to explore the dashboard to solve each task. The first task in each series will tell you if your answer is correct or not, but the second task won't provide feedback. Each tutorial and pair of tasks cover different features of the dashboard.

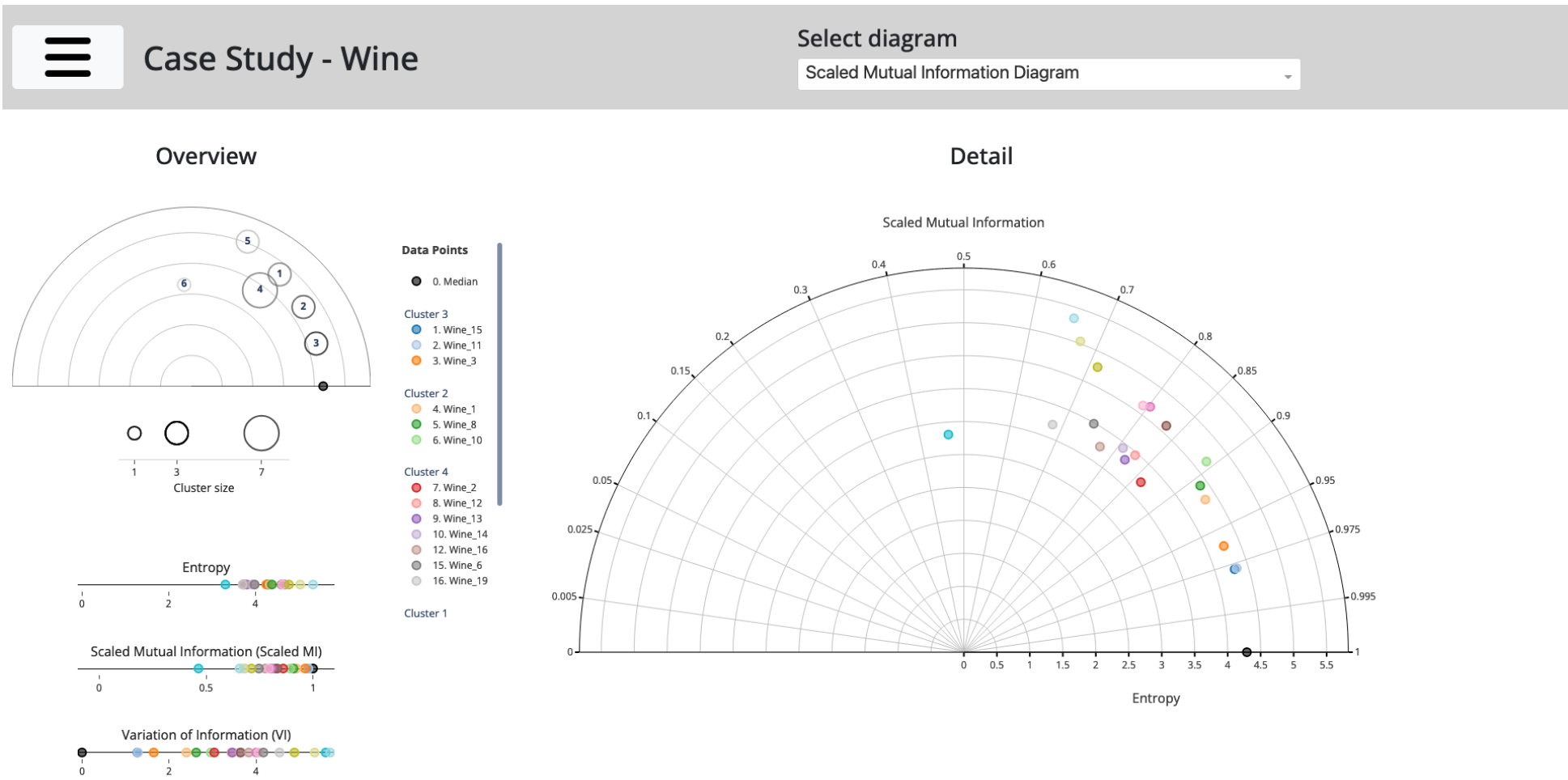
But first, let's prepare the dashboard and load the data.

Please open the dashboard using the following link: https://polar-diagrams-dashboard.onrender.com/. The dashboard will open in a new tab and might take some time to load. **Please be patient and keep this tab open for the whole duration of the questionnaire.**

You may need to wait a moment for the diagrams to update. Once the dashboard has fully loaded, your screen should match the image below. If this is the case, please confirm and continue.

You may encounter two problems:

- **The dashboard hasn't loaded properly**. If only some parts of the dashboard have loaded and the tab title doesn't say *Updating*, please refresh the tab.
- **The dashboard diagrams are one above the other**. If this is the case, please zoom out a bit (CTRL-) before they are arranged as in the image below.

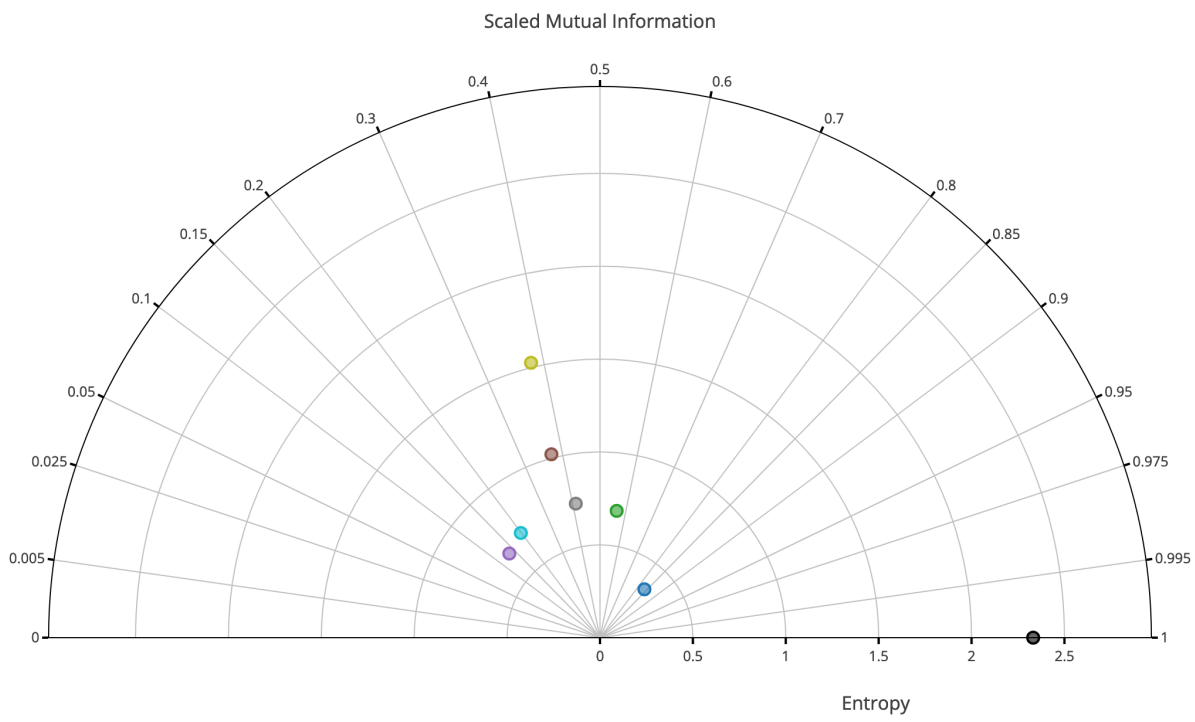


NA04

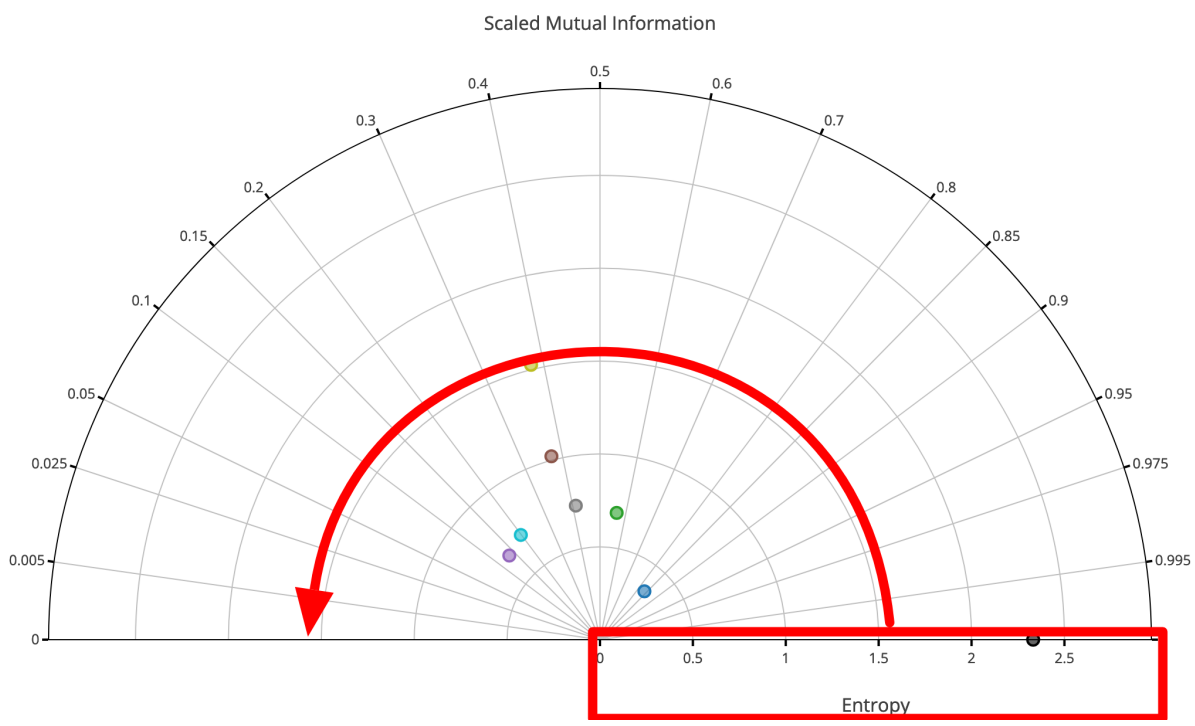
☐ I confirm that the dashboard in my other tab is as in the image above.

TS03

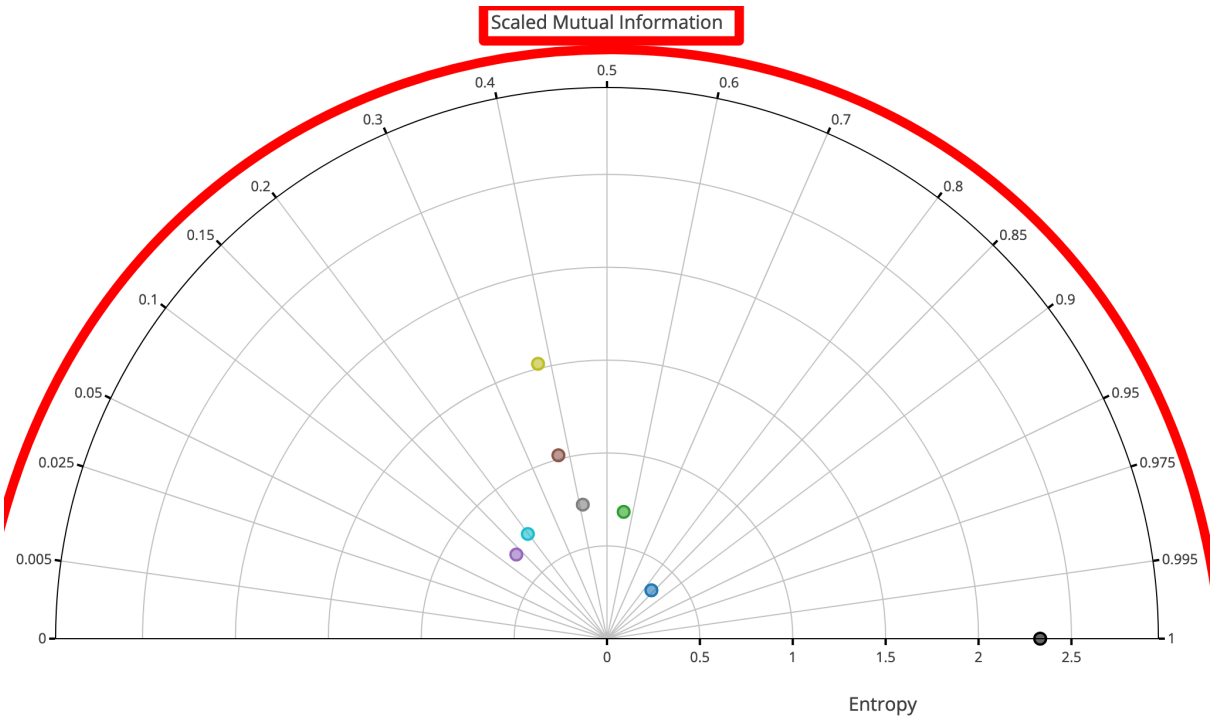
Tutorial 1



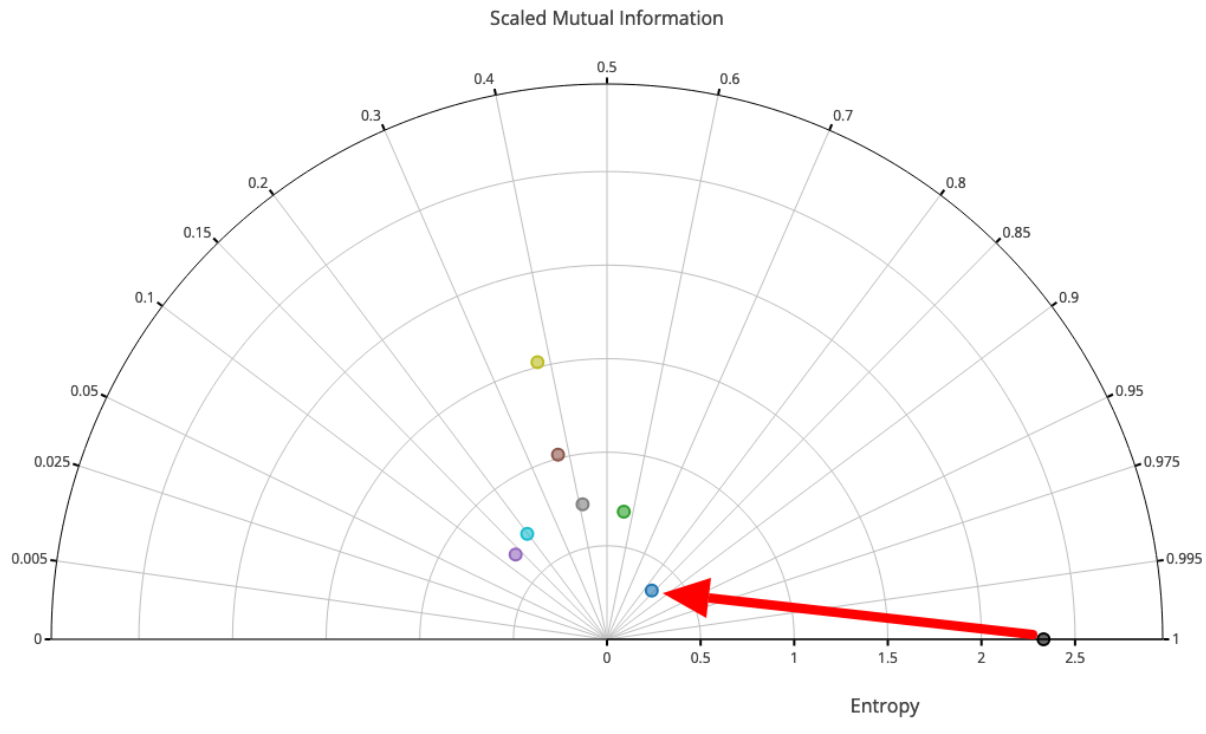
This is a scaled mutual information diagram. This polar diagram utilizes information-theoretic measures to visualize data points we call models (for traditional reasons). The black model, which is called the **reference model**, is always placed on the bottom axis.



The bottom or radial axis of the diagram depicts the amount of **entropy** contained in each model. Entropy represents the uncertainty within a model. The entropy axis spans the entire diagram, as shown in the image on the left.



The upper or angular axis of the diagram depicts the **scaled mutual information (scaled MI)** between the reference model and all other models. Scaled mutual information indicates how much we are surprised by a model, given our knowledge of the reference model.



The distance between the reference model and each other model depicts the **variation of information (VI)** between them. This measure indicates how similar every model is to the reference model. The closer a model is to the reference, the more similar they are, and the lower the variation of information between them.

In this image, we can see that the blue model is closest to the reference model.

Page 05

TASK1 E

TS04

9. Which model is closest to the reference model?

Please take a look at the other tab with the dashboard.

- ◯ Wine_15
- ◯ Wine_18
- ◯ Wine_7
- ◯ Wine_3
- ◯ Wine_2
- ◯ Wine_0

TS15

{EM_CROSS_MARK} This is wrong!

TS16

{EM_WHITE_HEAVY_CHECK_MARK} This is correct!

TS17

Page 06

TASK1 T

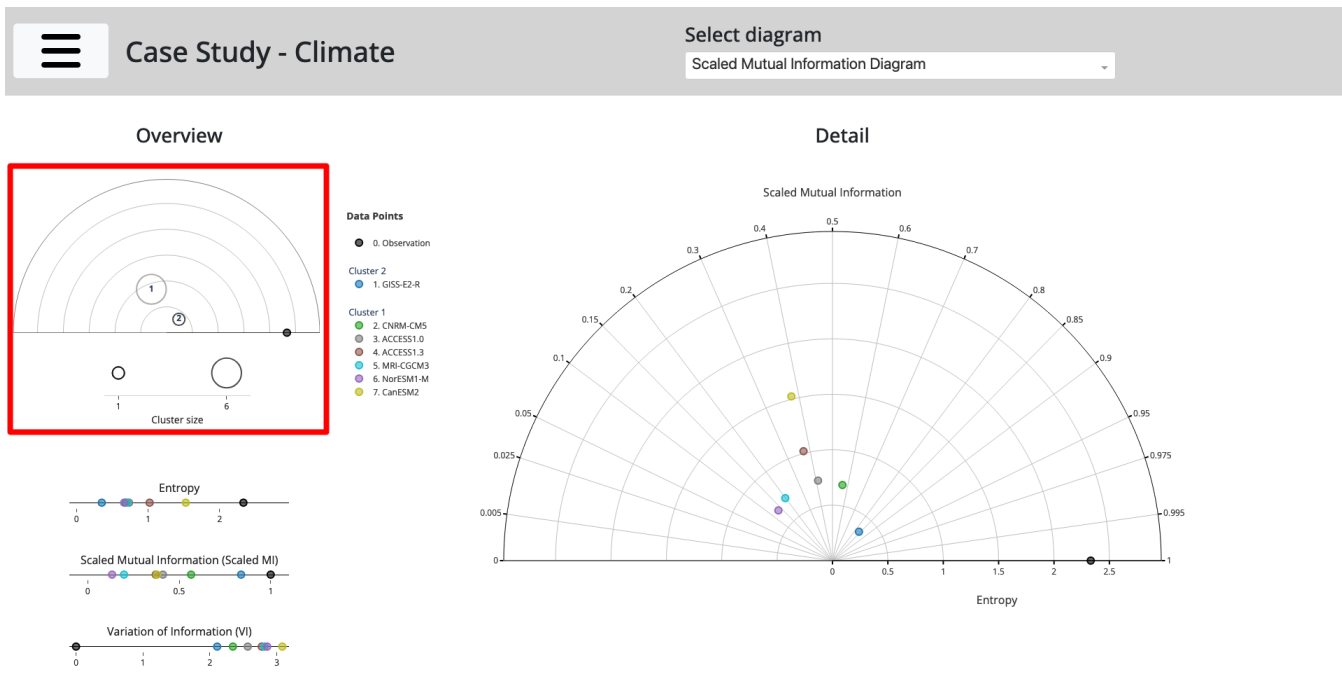
TS05

10. Which 3 models are the furthest from the reference model?

Please take a look at the other tab with the dashboard.

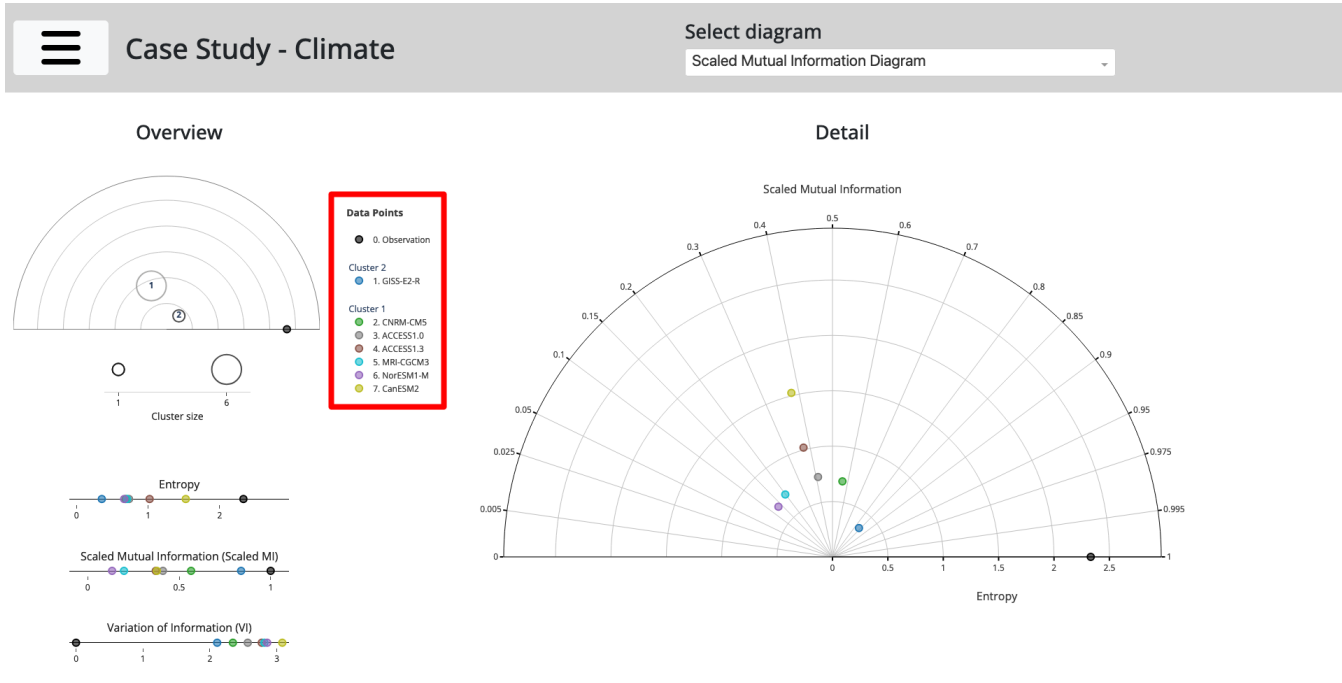
- ◯ Wine_15, Wine_11, and Wine_3
- ◯ Wine_18, Wine_17, and Wine_9
- ◯ Wine_19, Wine_6, and Wine_7
- ◯ Wine_0, Wine_5, and Wine_4
- ◯ Wine_18, Wine_13, and Wine_2
- ◯ Wine_17, Wine_9, and Wine_7

Tutorial 2



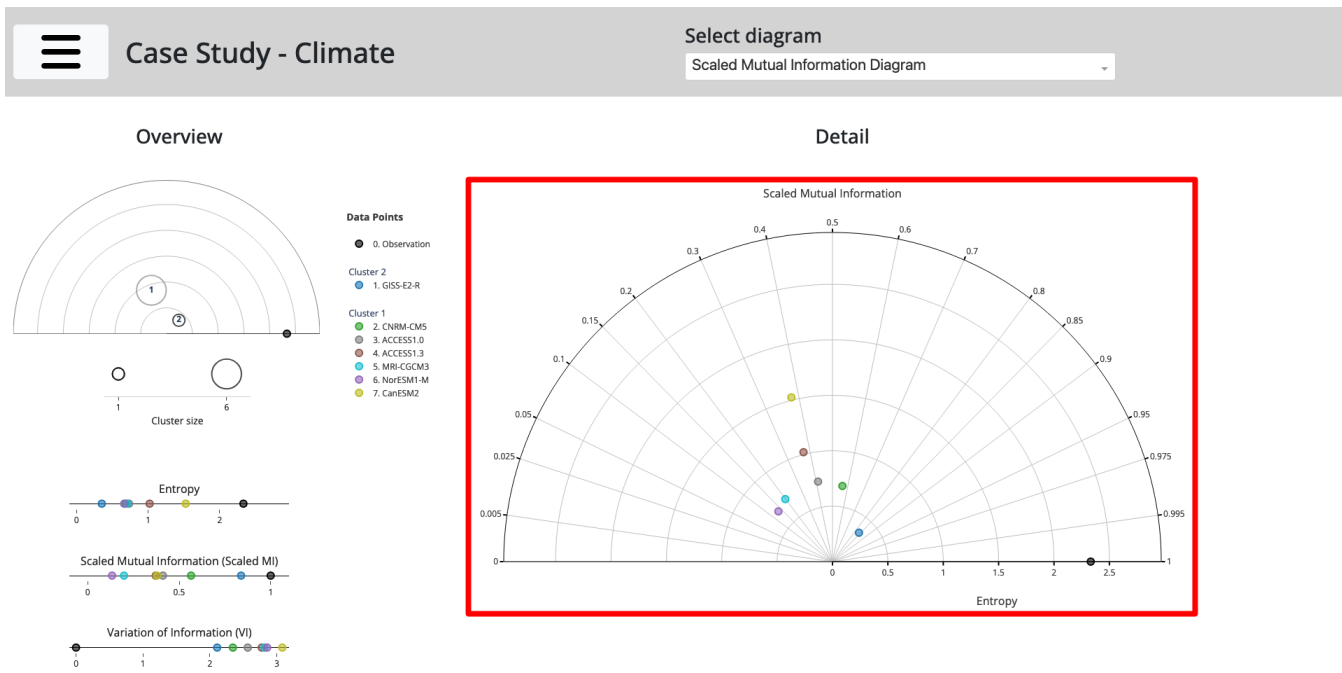
The leftmost part of the page provides an overview with a cluster legend below. The overview displays the same data as the detailed diagram on the right, but in a simplified, clustered view. The overview also enables interactive actions, which will be covered later.

In this image, we can see that we have two distinct clusters.



The central part of the page features the main legend, which links the overview and detailed views. This legend enables interactive actions through clicking or double-clicking its items (models). The legend items are sorted in decreasing order by their distance from the reference model. Additionally, the legend reflects the model clusters present in the detailed diagram and visualized in the overview.

In this image, we can see that the green model from Cluster 1 is the second closest to the reference model.



The rightmost part of the page presents a detailed view with a main legend on the left. This detailed view displays the data in greater detail. Hovering over the points in this diagram triggers a tooltip with additional information.

Page 08

TASK2 E

TS07

11. How many clusters are there, not counting the reference model?

Please take a look at the other tab with the dashboard.

- 5
- 0
- 7
- 6
- 2
- 3

TS15

{EM_CROSS_MARK} This is wrong!

TS16

{EM_WHITE_HEAVY_CHECK_MARK} This is correct!

TS18

Page 09

TASK2 T

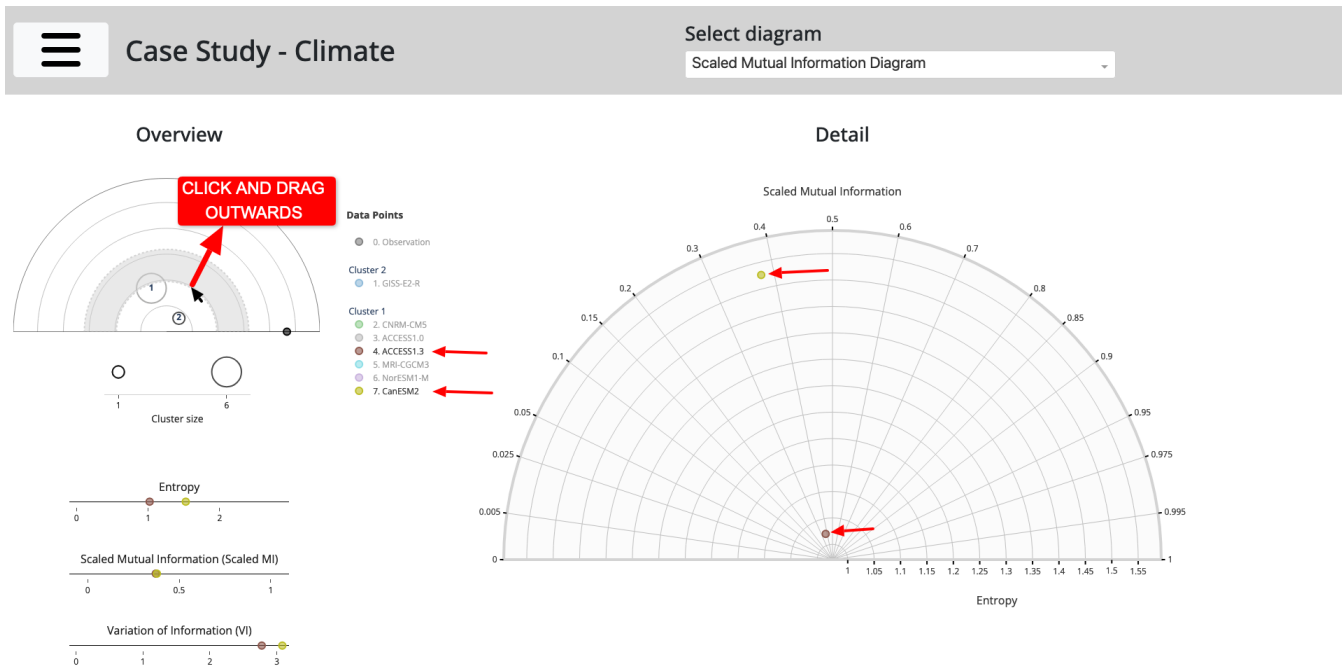
TS08

12. How many models are in Cluster 4?

Please take a look at the other tab with the dashboard.

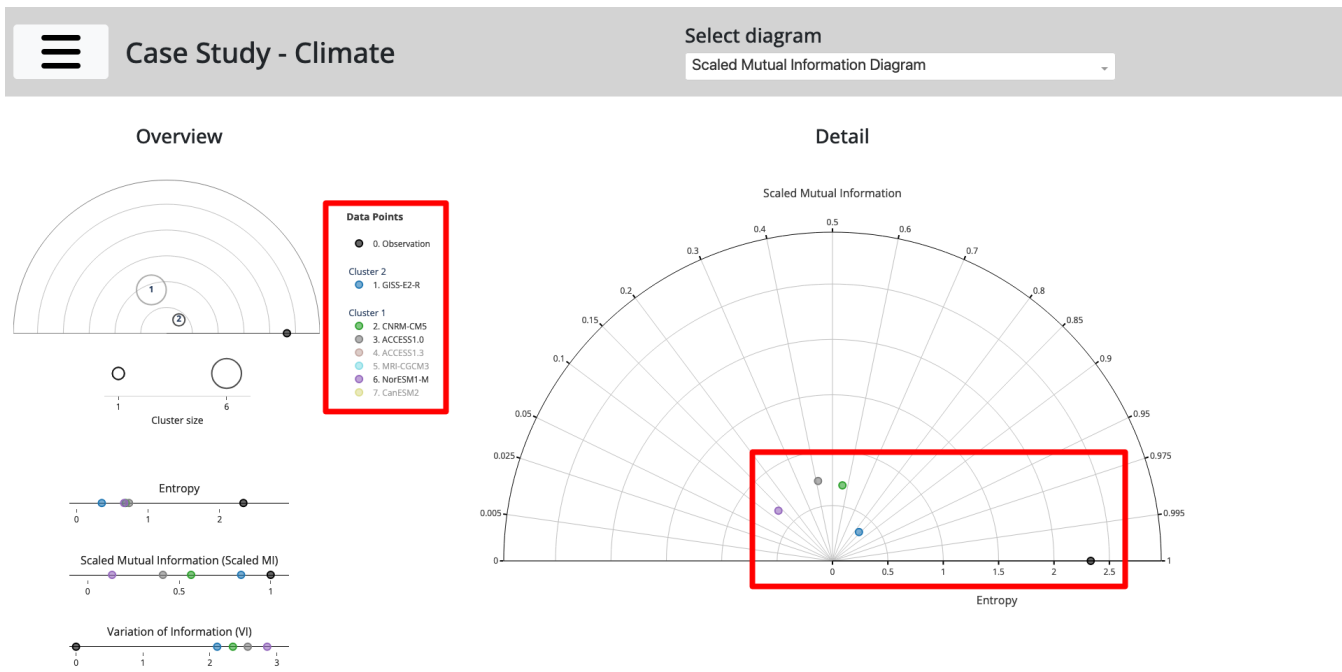
- 7
- 1
- 5
- 3
- 2
- 6

Tutorial 3



The overview diagram enables interactive brushing of the polar sectors (rings). Users can **click and drag** over the diagram to select a specific polar sector. The selection can be reset by double-clicking anywhere in the overview. This selection is reflected in the detailed diagram by adjusting the minimum and maximum axis values, coloring them in grey, and removing unselected data points. The selection is also highlighted in the main legend.

In this image, we can see that by brushing a part of the Cluster 1 we selected its two models in brown and yellow.



The main legend is interactive, enabling users to click or double-click its items. A **single click** selects or deselects an individual item. **Double-clicking** an item selects or deselects the entire cluster that item belongs to. Beware, double-clicking the cluster name (*e.g.* Cluster 1) does not have any effect. Once the user selects one cluster, a single click either deselects the items within that cluster or selects items outside the cluster. These changes are then reflected in the detailed diagram.

In this image, we can see that the selection of the blue, green, grey, and purple models is reflected in the detailed diagram.

Page 11

TASK3 E

TS10

13. Select Cluster 1 using the legend or overview. Which individual model at the center of that cluster has the middle/median variation of information (VI)?

Please take a look at the other tab with the dashboard.

- ◯ Wine_0
- ◯ Wine_14
- ◯ Wine_6
- ◯ Wine_10
- ◯ Wine_5
- ◯ Wine_4

TS15

{EM_CROSS_MARK} This is wrong!

TS16

{EM_WHITE_HEAVY_CHECK_MARK} This is correct!

TS19

Page 12

TASK3 T

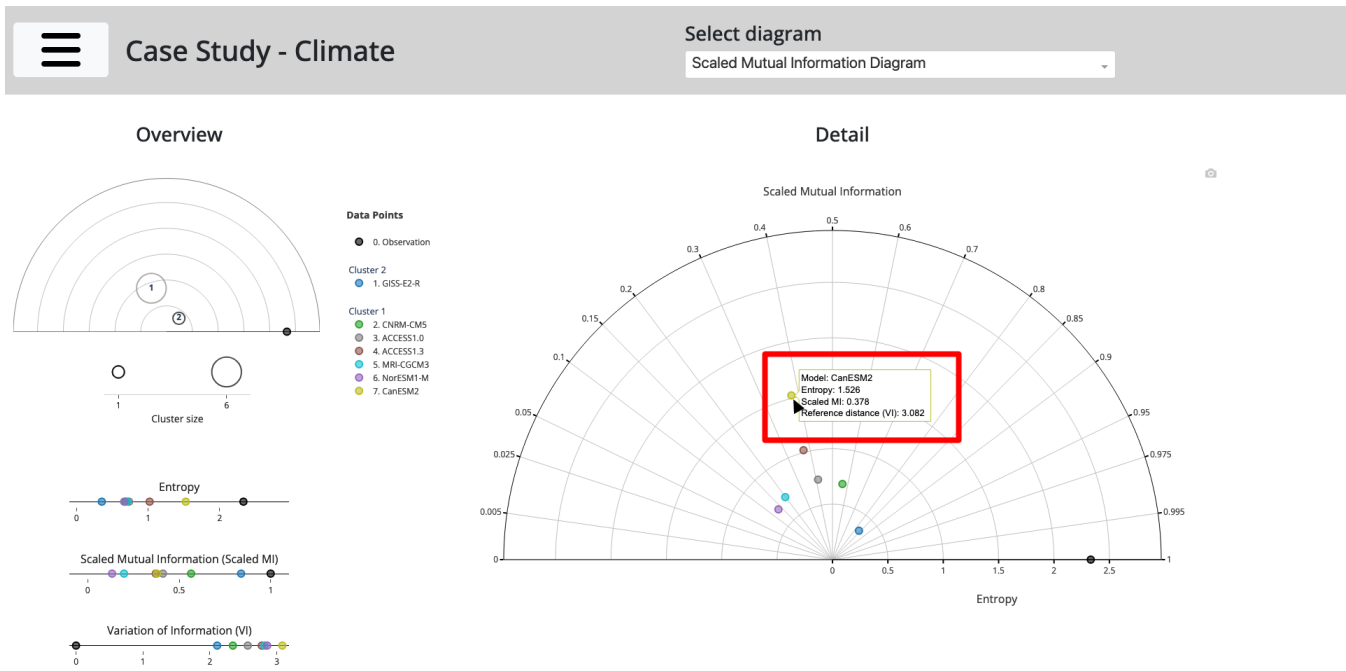
TS11

14. What model from Cluster 1 is furthest from Cluster 5?

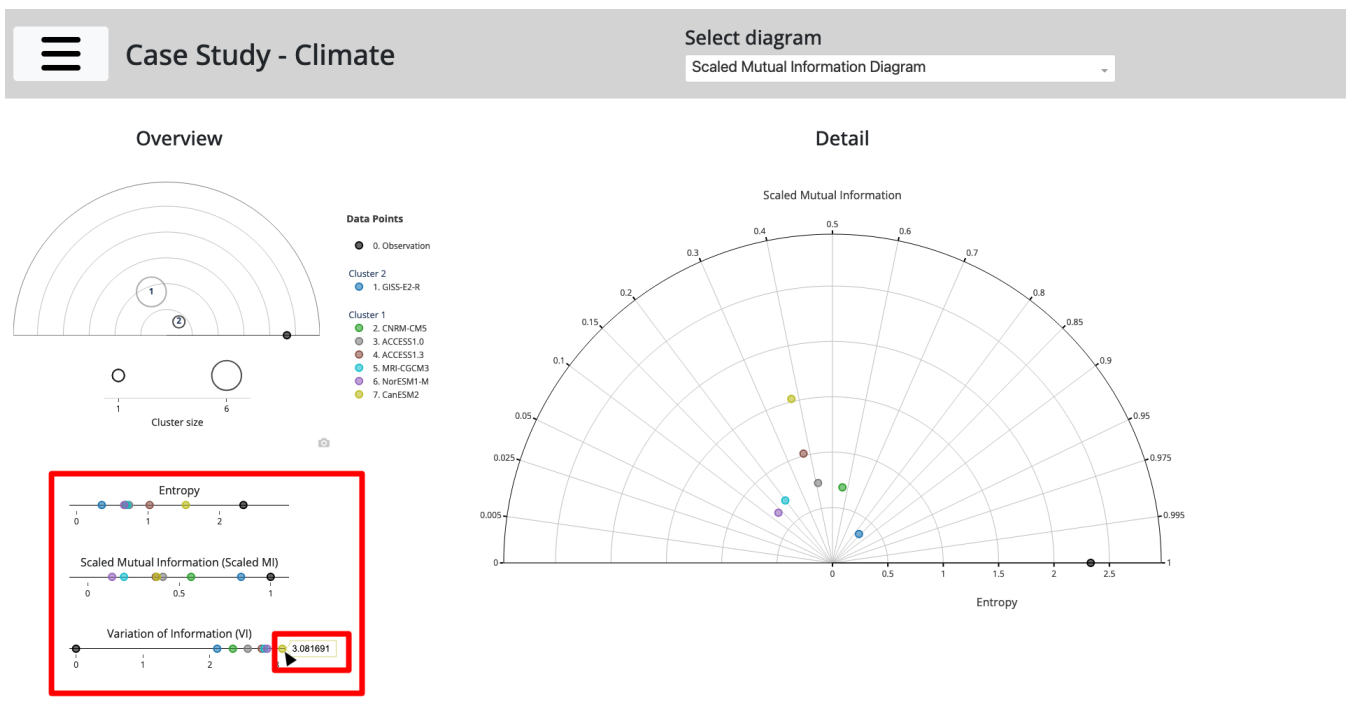
Please take a look at the other tab with the dashboard.

- ◯ Wine_5
- ◯ Wine_4
- ◯ Wine_0
- ◯ Wine_7
- ◯ Wine_10
- ◯ Wine_9

Tutorial 4



The detailed diagram implements a tooltip that appears when hovering over a model. The tooltip displays the name of the model and the exact values of each axis for the selected model. These values are also represented in the Cartesian-linking plot shown below.



The final part of the page is a Cartesian-linking plot that simplifies the detailed diagram further. Each line in this plot represents one axis of the detailed diagram, with the final axis (named Variation of Information) depicting the distance between the reference model and every other model. Each line also features a tooltip that displays the exact value for that axis across all models.

Page 14

TASK4 E

TS13

15. Select models Wine_4 and Wine_17. Check the tooltip and Cartesian-linking plot. What is the approximate sum of their entropies?

Please take a look at the other tab with the dashboard.

- ◯ Approximately 1
- ◯ Approximately 15
- ◯ Approximately 12
- ◯ Approximately 10
- ◯ Approximately 4
- ◯ Approximately 8

TS15

{EM_CROSS_MARK} This is wrong!

TS16

{EM_WHITE_HEAVY_CHECK_MARK} This is correct!

TS20

Page 15

TASK4 T

TS14

16. Select Cluster 3 and Cluster 5. What are the ranges of variation of information (VI) within these two clusters? Provide the minimum and maximum VI values for these clusters, starting from smallest to largest.

Please take a look at the other tab with the dashboard.

- ◯ 2.3 – 2.9 and 4.8 – 5.7
- ◯ 3.0 – 4.5 and 3.6 – 4.0
- ◯ 1.2 – 1.6 and 3.6 – 4.0
- ◯ 2.5 – 3.2 and 4.2 – 5.2
- ◯ 0.5 – 2.2 and 3.6 – 4.0
- ◯ 1.2 – 1.6 and 4.8 – 5.7

OU02

Thank you!

Thank you for participating in the questionnaire! Your input is incredibly valuable and greatly appreciated.

Before you leave, please answer one more question about the usability of the dashboard you've been using. This feedback will help us improve it further.

After that, you have the option to leave any comments or suggestions about the questionnaire, dashboard, diagrams, or anything else that comes to mind. Feel free to leave this section blank if you don't have any additional feedback.

If you would like to be acknowledged in the published article, please send an email to AnzelA@rki.de with the subject line "**Request for Acknowledgement**".

OU01

17. How would you rate the online web application you have been working with?

	Strongly disagree				Strongly agree
I think that I would like to use this system frequently.	○	○	○	○	○
I found the system unnecessarily complex.	○	○	○	○	○
I thought the system was easy to use.	○	○	○	○	○
I think that I would need the support of a technical person to be able to use this system.	○	○	○	○	○
I found the various functions in this system were well integrated.	○	○	○	○	○
I thought there was too much inconsistency in this system.	○	○	○	○	○
I would imagine that most people would learn to use this system very quickly.	○	○	○	○	○
I found the system very cumbersome to use.	○	○	○	○	○
I felt very confident using the system.	○	○	○	○	○
I needed to learn a lot of things before I could get going with this system.	○	○	○	○	○

OU03

18. Feel free to provide feedback about this questionnaire, dashboard, diagrams, or anything else that comes to mind.

The answer to this question is optional.

Last Page

Thank you for completing this questionnaire!

We would like to thank you very much for helping us.

Your answers were transmitted, you may close the browser window or tab now.

Dr. Aleksandar Anžel, Robert Koch Institute, ZKI-PH 5 – 2024